Prof. Olga's Class - September 8, 2025

2025-09-05

Dear prof. Olga. How are you? I hope i did a good job in this task. It was challenging, because I've missed the R workshop, but I'll try my best.

1 - How many Korean Dramas are included in this list? Write an R code to respond to this question.

```
db_kdrama <- read.csv("kdrama.csv") #Open K-Drama database
sum(!is.na(db_kdrama$Name)) #Count the arrows in the "Name" colum
```

```
## [1] 250
```

```
#I also used !is.na ro eliminate NA values
```

2 - Use an R code to return a list of variables that are included in this dataset.

```
names(db_kdrama) #Return the name of the variables in the database
```

```
##  [1] "Name"                 "Aired.Date"           "Year.of.release"
##  [4] "Original.Network"     "Aired.On"             "Number.of.Episodes"
##  [7] "Duration"             "Content.Rating"       "Rating"
## [10] "Synopsis"             "Genre"                "Tags"
## [13] "Director"             "Screenwriter"         "Cast"
## [16] "Production.companies" "Rank"
```

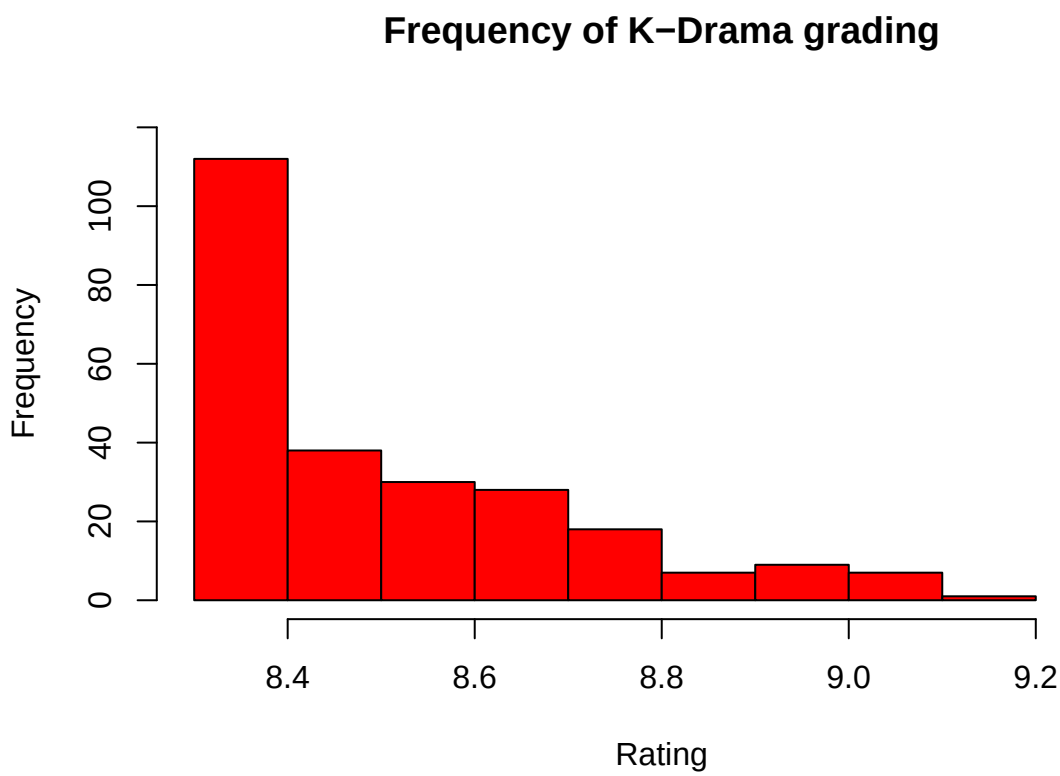
3 - Use an R code to identify what the mean value of total number of episodes for all the kdramas in the dataset is.

```
mean(db_kdrama$Number.of.Episodes)
```

```
## [1] 19.064
```

4 - Plot a histogram of the shows rating.

```
hist(db_kdrama$Rating, xlim = c(8.3, 9.3), ylim = c(0, 120),
       xlab = 'Rating', ylab = 'Frequency',
       main = 'Frequency of K-Drama grading',
       col = 'red', border = "black")
```



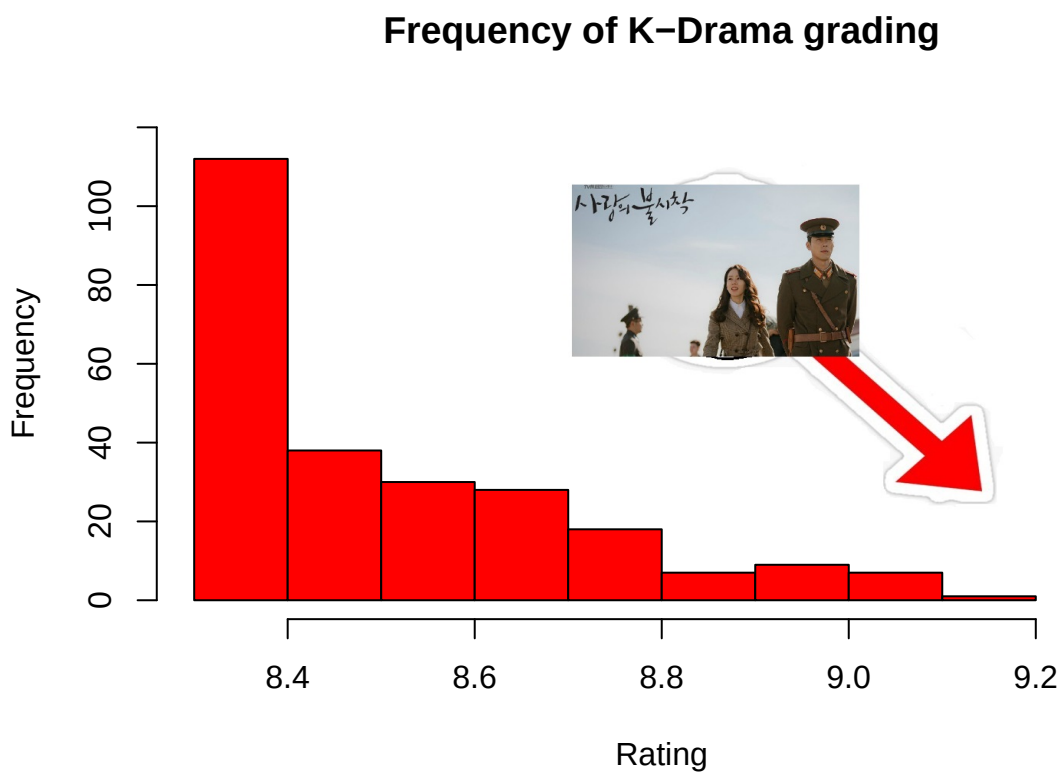

```
# I used hist to produce the histogram, xlim and ylim to define de limit values of the axis
# xlab and ylab to change labels, main to change the name of the histogram
# col and border to change colors
```

We can personalize the histogram in interesting ways, such as:

```
#First, install the jpeg package
install.packages("jpeg")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```
#Run the package
library(jpeg)
#Criate the object for an image (I made it for the K-drama "Crash Landing on You" or
#"Sarangui Bulsichak")
pousandonoamor <- readJPEG("chuudoloona.jpg")
#Then, create an empty plot
plot(1, type = "n", xlim = c(0, 1), ylim = c(0, 1), axes = FALSE, xlab = "", ylab = "")
#Add the image to the plot
rasterImage(pousandonoamor, xleft = 0, ybottom = 0, xright = 1, ytop = 1)
#Allow plot over image
par(new = TRUE)
#Plot the original histogram
hist(db_kdrama$Rating, xlim = c(8.3, 9.3), ylim = c(0, 120),
     xlab = 'Rating', ylab = 'Frequency', main = 'Frequency of K-Drama grading',
     col = 'red', border = "black") # Plot your histogram
```



5 - How many shows have a rating higher than 9 points? Show how you calculated that in R. There are different ways to calculate that in R.

```
#we can use the following code to return how many show have more than 9
sum(db_kdrama$Rating > 9) #excludes shows with exactly 9 rating
```

```
## [1] 8
```

```
sum(db_kdrama$Rating >= 9) #counts shows with 9 and more than 9
```

```
## [1] 17
```

```
# you can also use dplyr to create a database only with the shows rated greater than R
install.packages("dplyr") #install and load dplyr package
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
#we can use filter to create de database with shows rated 9 or greater
kdrama_g9 <- filter(db_kdrama, Rating >= 9)
dplyr::glimpse(kdrama_g9) #used to see the new database
```

```
## Rows: 17
## Columns: 17
## $ Name                 <chr> "Move to Heaven", "Flower of Evil", "Hospital Pla~
## $ Aired.Date           <chr> "May 14, 2021", "Jul 29, 2020 - Sep 23, 2020", "M~
## $ Year.of.release      <int> 2021, 2020, 2020, 2021, 2018, 2015, 2022, 2017, 2~
## $ Original.Network     <chr> "Netflix", "tvN", "Netflix,  tvN ", "Netflix,  tv~
## $ Aired.On             <chr> "Friday", " Wednesday, Thursday", "Thursday", "Th~
## $ Number.of.Episodes   <int> 10, 16, 12, 12, 16, 20, 8, 16, 20, 16, 20, 16, 16~
## $ Duration             <chr> "52 min.", "1 hr. 10 min.", "1 hr. 30 min.", "1 h~
## $ Content.Rating       <chr> "18+ Restricted (violence & profanity)", "15+ - T~
## $ Rating               <dbl> 9.2, 9.1, 9.1, 9.1, 9.1, 9.1, 9.1, 9.1, 9.0, 9.0,~
## $ Synopsis             <chr> "Geu Roo is a young autistic man. He works for hi~
## $ Genre                <chr> "Life,  Drama,  Family ", "Thriller,  Romance,  C~
## $ Tags                 <chr> "Autism, Uncle-Nephew Relationship, Death, Savant~
## $ Director             <chr> "Kim Sung Ho", "Kim Chul Gyu, Yoon Jong Ho", "Shi~
## $ Screenwriter         <chr> "Yoon Ji Ryun", "Yoo Jung Hee", "Lee Woo Jung", "~
## $ Cast                 <chr> "Lee Je Hoon, Tang Jun Sang, Hong Seung Hee, Jung~
## $ Production.companies <chr> "Page One Film, Number Three Pictures", "Monster ~
## $ Rank                 <chr> "#1", "#2", "#3", "#4", "#5", "#6", "#7", "#8", "~
```

```r
# I don't know why View function is not working, so let's use this code

#Using the new database, we can discover the grade of "Crash Landing on You"
crash_row <- kdrama_g9 %>% filter(Name == "Crash Landing on You")
  crash_rating <- crash_row$Rating
  print(crash_rating)
```

```
## [1] 9
```

It's rated 9, I almost guessed it right!

6 - Rename variable Year.of.release to simply Year without creating a new variable

```r
newdb_kdrama <- db_kdrama %>%
  rename(Year = Year.of.release)
names(newdb_kdrama)
```

```
##  [1] "Name"                 "Aired.Date"           "Year"
##  [4] "Original.Network"     "Aired.On"             "Number.of.Episodes"
##  [7] "Duration"             "Content.Rating"       "Rating"
## [10] "Synopsis"             "Genre"                "Tags"
## [13] "Director"             "Screenwriter"         "Cast"
## [16] "Production.companies" "Rank"
```

7 - How many shows in this dataset were released in 2020-2022? Demonstrate with an R code.

```r
kdrama_2022 <- db_kdrama %>%
  filter(Year.of.release >= 2020 & Year.of.release <= 2022)
  dim(kdrama_2022)
```

```
## [1] 106  17
```

106 shows were released in the 2020-2022 period!

8 - What type does the variable Duration have? Show with an R code.

```r
class(db_kdrama$Duration)
```

```
## [1] "character"
```

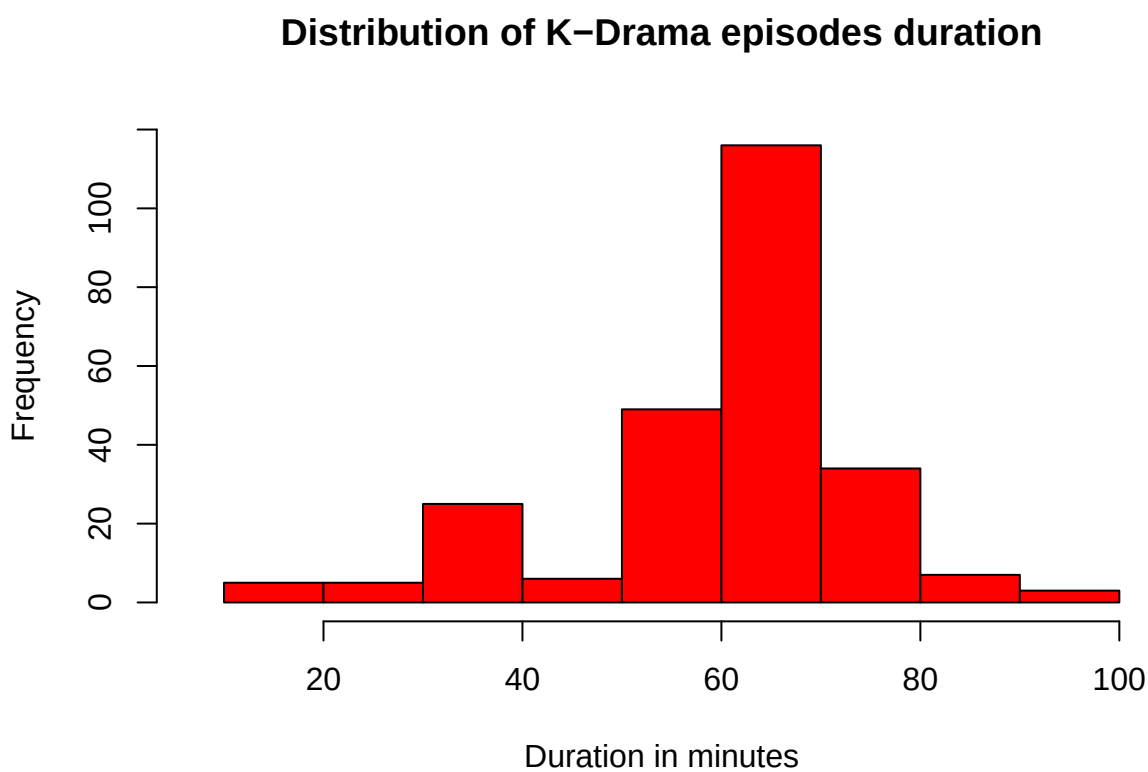
9 - Recode variable Duration so that it would be a numerical variable measuring duration in minutes. Plot the histogram of the recoded variable (though one).

```
# We will need to install and load "stringr" package
install.packages("stringr")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```
library(stringr)
# I'll use the function str_extract to remove the "hr." and "min." from observations
# I'll use is.na to check whether hours or minutes are absent, and change it for zero
# Extract hours, returns NA if not found
hours <- as.numeric(str_extract(db_kdrama$Duration, "\\d+(?= hr\\.)"))
# Turns NAs into 0
hours[is.na(hours)] <- 0
# Extract minutes, returns NA if not found
minutes <- as.numeric(str_extract(db_kdrama$Duration, "\\d+(?= min\\.)"))
# Turns NAs into 0
minutes[is.na(minutes)] <- 0
# Turns everything into minutes
total_min <- hours * 60 + minutes

#Plot the data into a histogram
hist(total_min, xlim = c(7, 101), ylim = c(0, 120),
     xlab = 'Duration in minutes', ylab = 'Frequency',
     main = 'Distribution of K-Drama episodes duration',
     col = 'red', border = "black")
```



10 - Create a new dataset that will include shows with Original.Network being Netflix.

```r
# I use str_detect to find Netflix in observations that include other networks
netflix_shows <- db_kdrama %>%
    filter(str_detect(Original.Network, "Netflix"))
```

11 - What is the average rating score for the shows that have Netflix as an Original Network.

```r
mean(netflix_shows$Rating)
```

```
## [1] 8.6625
```